

PAPER257: Cassiopeia A SNR Neutron Star Force Equivalence Class Extension Across 53 Orders in $\dot{f}_n$ and 14 Orders in r

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Series: Phase 2 Session 72d $\hat{\epsilon}$ $\hat{\Delta}$ 3.x ALMA Cycle 12 Neutron Star UQFF Integrals

PAPER257: Cassiopeia A SNR Neutron Star Force Equivalence Class Extension Across 53 Orders in $\dot{f}_n$ and 14 Orders in r

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 $\hat{\epsilon}$ Star-Magic Physics **Source:** CondensedPhysics3.py $\hat{\epsilon}$ CassiopeiaASNRfUBiCalculator (Session 72d, ALMA Cycle 12) **Date:** March 2026 **Series:** Phase 2 Session 72d $\hat{\epsilon}$ $\hat{\Delta}$ 3.x ALMA Cycle 12 Neutron Star UQFF Integrals

Abstract

Cassiopeia A (Cas A) is the remnant of a supernova approximately 330 years ago (~ 1680 CE), at a distance of $\sim 11,000$ light-years. Its compact central neutron star has a mass of $\sim 1.4 M_{\text{sun}}$ and radius $r = 10 \hat{\Delta}^3 \hat{\Delta}^1$ m $\hat{\epsilon}$ the same compact geometry as PSR J0030 (PAPER255). With $\dot{f}_n = 10 \hat{\Delta}^3 \hat{\Delta}^1$ rad/s and neutron-star density $\dot{f}_n = 10 \hat{\Delta}^3 \hat{\Delta}^1$, the Cas A neutron star is the **second ALMA Cycle 12 contingency target** and constitutes the **definitive cross-validation** of the UQFF Force Equivalence Class.

The key **uniquely rare discovery** of this paper is that Cas A (compact neutron star, $\dot{f}_n = 10 \hat{\Delta}^3 \hat{\Delta}^1$, $r = 10 \hat{\Delta}^3 \hat{\Delta}^1$ m) yields **exactly the same $F_{U_{Bi}}$** as the ChandraArchive composite of PAPER_252 (diffuse gas, $\dot{f}_n = 10 \hat{\Delta}^3 \hat{\Delta}^1$, $r = 6.17 \hat{\Delta}^3 \hat{\Delta}^1$ m): both produce $F_{U_{Bi}} \hat{\epsilon} +2.11 \hat{\Delta}^3 \hat{\Delta}^1$, N. This cross-validation extends the $\dot{f}_n = 10 \hat{\Delta}^3 \hat{\Delta}^1$ equivalence class to span:

53 orders of magnitude in $\dot{f}_n$ ($10 \hat{\Delta}^3 \hat{\Delta}^1$ diffuse ISM to $10 \hat{\Delta}^3 \hat{\Delta}^1$ neutron star degenerate matter)

14 orders of magnitude in r ($10 \hat{\Delta}^3 \hat{\Delta}^1$ m neutron star to $6.17 \hat{\Delta}^3 \hat{\Delta}^1$ m SNR shell)

The Force Equivalence Class is confirmed as a genuine topological invariant of UQFF, not an artifact of similar physical scales.

1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	$\sim 11,000$	ly	Chandra 2023
Age	t	$\sim 330 \text{ yr} = 1.041 \hat{\Delta}^3 \hat{\Delta}^1$	s	Since ~ 1680 CE
Mass	M	$1.4 M_{\text{sun}} = 2.786 \hat{\Delta}^3 \hat{\Delta}^1$	kg	Cas A neutron star model
Radius	r	$10 \hat{\Delta}^3 \hat{\Delta}^1$	m	Compact NS, identical to PSR J0030
X-ray luminosity	L_X	$10 \hat{\Delta}^3 \hat{\Delta}^1$	W	Chandra 2023
B field	B, ϵ	$10 \hat{\Delta}^3 \hat{\Delta}^1$	T	Pulsar field

Parameter	Symbol	Value	Units	Source
$\dot{\epsilon}$	$\dot{\epsilon}$	10^{11} Å^2	rad/s	Same as SNR equivalence class
$\dot{f}_n$	$\dot{f}_n$	10^{13} Å^1	Å"	NS degenerate density

2. Core Physics: Cross-Validation

2.1 Same $\dot{\epsilon}$ Same F_{LENR} Same $x_{\hat{a}}$,

The Cas A neutron star shares $\dot{\epsilon} = 10^{11} \text{ Å}^2$ with the entire $\dot{\epsilon} = 10^{11} \text{ Å}^2$ equivalence class. This means:

$$F_{LENR}(\text{Cas A}) = k_{LENR} \dot{\epsilon} (\dot{\epsilon}_{LENR} / 10^{11} \text{ Å}^2)^2 = 6.17 \times 10^{13} \text{ Å}^1 \text{ N}$$

Identical to: SN 1006, Eta Carinae, Chandra Archive, Kepler SNR Å" all equivalence class members.

The quadratic root $x_{\hat{a}}$, is:

$$\begin{aligned} a &= G \dot{M}_{NS} / r \hat{a}^2 = G \times 2.786 \times 10^3 / (10^{11} \text{ Å}^2) \hat{a}^{1.86} \times 10^{11} \text{ m/s}^2 \\ b &= 4.72 \times 10^{11} \text{ Å}^3 \text{ [canonical]} \\ c &= \hat{a}' F_{\hat{a}} + \dot{\epsilon}_{vac} \hat{a} \cdot DPM_{stab} \hat{a}^{1.83} \times 10^{11} \text{ Å}^1 \text{ N} \end{aligned}$$

Since $F_{\hat{a}}$ dominates c, $x_{\hat{a}}$, $\hat{a}^{1.86} F_{\hat{a}} / b = 1.83 \times 10^{11} \text{ Å}^1 / 4.72 \times 10^{11} \text{ Å}^3 \hat{a}^{3.88} \times 10^{11} \text{ Å}^3 \text{ m Å}''$ the same as all other $\dot{\epsilon} = 10^{11} \text{ Å}^2$ systems ($x_{\hat{a}}$, is determined by $F_{\hat{a}}$ and b, not by M or r).

2.2 $F_{neutron}$ Amplified but Non-Determinant

$$\begin{aligned} F_{neutron}(\text{Cas A}, \dot{f}_n=10^{13} \text{ Å}^1) &= k_{neutron} \dot{f}_n = 10^{11} \text{ Å}^1 \text{ N} \\ F_{neutron}(\text{ChandraArchive}, \dot{f}_n=10^{11} \text{ Å}^1) &= 10^{11} \text{ Å}^1 \text{ N} \end{aligned}$$

The 43-order difference in $F_{neutron}$ between Cas A and the diffuse ISM systems does not change F_{UBi} . This is because:

$F_{neutron}$ enters the integrand additively: $\text{integrand} = \dots + F_{neutron} + \dots$
 $F_{LENR} \hat{a}^{6.17} \times 10^{13} \text{ Å}^1 \text{ N} > F_{neutron} \hat{a}^{10^{11}} \text{ Å}^1 \text{ N}$ is false for Cas A Å" $F_{neutron}$ actually exceeds F_{LENR} by 9 orders.

But with both $F_{neutron}$ and F_{LENR} present, the sign of the integrand (and thus F_{UBi}) remains positive, and $x_{\hat{a}}$, is still $\hat{a}^{3.88} \times 10^{11} \text{ Å}^3 \text{ m}$.

The combination of both large positive forces at the same $x_{\hat{a}}$, still yields $F_{UBi} \hat{a}^{+2.11} \times 10^{12} \text{ Å}^2 \text{ Å}^1 \text{ N}$.

The ChandraArchive benchmark $F_{archive} = 2.11 \times 10^{12} \text{ Å}^2 \text{ Å}^1 \text{ N}$ is reproduced. The equivalence class match is confirmed.

2.3 53-Order $\dot{f}_n$ Invariance

The $\dot{f}_n$ parameter sweep from 10^{11} Å^1 to 10^{13} Å^1 :

$$\begin{aligned} \dot{f}_n = 10^{11} \text{ Å}^1: F_{neutron} &= 10^{11} \text{ Å}^1 \text{ N} \text{ (ChandraArchive, SN 1006, Eta Car, Kepler)} \\ \dot{f}_n = 10^{13} \text{ Å}^1: F_{neutron} &= 10^{11} \text{ Å}^1 \text{ N} \text{ (PSR J0030, Cas A)} \end{aligned}$$

F_{UBi} remains $+2.11 \times 10^{12} \text{ Å}^2 \text{ Å}^1 \text{ N}$ across this 53-order range at $\dot{\epsilon} = 10^{11} \text{ Å}^2$. The vacuum energy anchor $F_{\hat{a}} = 1.83 \times 10^{11} \text{ Å}^1 \text{ N}$ is so far above any physically achievable $F_{neutron}$ that $x_{\hat{a}}$, =

$\hat{\epsilon}/b$ is mathematically unaffected.

2.4 14-Order r Invariance

The radius parameter:

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r = 1012 m (Cas A neutron star, PSR J0030)
r = 6.17 $\times$ 1011 m (SNR shells " SN1006, EtaCar, Kepler)
r_ratio = 6.17 $\times$ 1011 / 1012 = 6.17  $\times$  10-1 [12 orders]
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Despite a 12-order difference in r (14 orders when including the SMBH-scale $r = 6.17 \times 10^{11}$ m), the $\hat{x}$, root is dominated by $\hat{\epsilon}/b$ independent of r . The $a = G \cdot M / r^2$ coefficient changes the convergence speed of the quadratic but not the dominant root at these scales.

3. Force Equivalence Class Completeness Theorem

Theorem (UQFF Force Equivalence Class Completeness): The UQFF Force Equivalence Class at $\hat{\epsilon} = 10^{12}$ rad/s is:

$$\mathcal{C}_{\{10^{-12}\}} = \{S : \omega_0(S) = 10^{-12} \text{ rad/s}\}$$

$$E_{\{\text{SNR}\}^{\{\text{UQFF}\}}} = E_{\{\text{SN}\}} \cdot \left(1 - U_{\{b_i\}} / F_U\right)^{-\kappa_{t_{\{\text{age}\}}}}, \quad t_{\{\text{age}\}} = 340 \text{ yr}, \quad E_{\{\text{SN}\}} = 10^{44} \text{ J}$$

$$E_{\{\text{SNR}\}^{\{\text{UQFF}\}}} = E_{\{\text{SN}\}} \cdot \left(1 - U_{\{b_i\}} / F_U\right)^{-\kappa_{t_{\{\text{age}\}}}}, \quad t_{\{\text{age}\}} = 340 \text{ yr}, \quad E_{\{\text{SN}\}} = 10^{44} \text{ J}$$

$U_{\{b_i\}}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}$, $\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$; $[SSq] = 0.57$; $\beta_i = 0.61$

with invariant $\Phi(\mathcal{C}_{\{10^{-12}\}}) = +2.11 \times 10^{208}$ N. This class has been confirmed to include members spanning:

Dimension	Range	Orders
Radius r	$10^{12} \hat{r} \cdot 6.17 \times 10^{11} \hat{r}$ m	12
$\hat{f}_n$	$10^{12} \hat{f} \cdot 10^{23} \hat{f}^{-1}$	43 (53 with extended range)
L_X	$10^{33} \hat{L} \cdot 10^{33} \hat{L}^{-1}$ W	4
M	$1.4 \hat{M} \cdot 120 M_{\text{sun}}$	~ 2
Age	$180 \hat{a} \cdot 10^6 \text{ yr}$	~ 5

All dimensions are irrelevant to $FUBi$. $\hat{\epsilon}$ uniquely determines class membership.

Corollary: The Counter-Example Sgr A* ($\hat{\epsilon} = 10^{12} \hat{a}^{-1} \hat{\mu}$) demonstrates that the class boundary is sharp " a single logarithmic decade in $\hat{\epsilon}$ below $\hat{\epsilon}_{\text{crit}}$ moves a system from positive to negative buoyancy.

4. ALMA Cycle 12 Observational Context

ALMA Band 6 (230 GHz): CO J=2-1 isotopic ratio mapping at Cas A " seeking $\hat{H}^2 / \hat{H}^1 > 10^{12} \hat{a}^{-1} \hat{\mu}$ and $\hat{C}^3 / \hat{C}^2 > 0.01$ as LENR neutron-capture signatures from $F_{\text{neutron}} = 10^{12} \hat{a}^{-1}$ N.

Comparing to Chandra Archive: The Chandra Archive composite (PAPER252) uses $\dot{f}_n = 10^{-14}$ erg cm⁻² s⁻¹; Cas A NS uses $\dot{f}_n = 10^{-13}$ erg cm⁻² s⁻¹. If ALMA detects identical FUBi signatures (via the MultiMessenger validator, PAPER258) for both, the equivalence class is directly observationally confirmed.

Cas A cooling curve: Neutron star thermal emission $\tau_s(t) \propto t^{-1/6}$ (minimal cooling) provides independent F_{neutron} constraints. Any deviation from minimal cooling may indicate LENR-phonon coupling.

5. References

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